Annexure -C

**GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR**
**Department of Computer Science & Engineering**
**Fortnightly Review Project Progress Review Report starting from December 6th ,2025**

**Student Group Details**

| S. No. | Roll Number | Registration Number | Student Name |
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**Project Information**

Project Title: Image Based Cattle Measurement and Trait Estimation System

Domain / Area: Computer Vision , AI

Faculty Supervisor: Mr. V. S. K. Chaitanya

**Tasks and Progress**

| Tasks Planned | Tasks Completed | Remarks / Challenges |
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| **1. Problem Statement & Objectives**<br><br>1.1 Briefly explain the problem statement. | **1.1 Problem Statement:**<br>Manual Animal Type Classification, despite being conducted by trained animal typers, is inherently subjective and susceptible to errors due to fatigue, visual bias, and measurement inaccuracies. These inconsistencies affect data quality and can lead to suboptimal selection of elite dams under national breeding programs such as the Rashtriya Gokul Mission. With large-scale data collection required across diverse field conditions, there is a critical need for an automated, objective, and scalable system that can reliably evaluate animal body structure and assign standardized ATC scores with minimal human intervention. | Although the problem of subjectivity in manual Animal Type Classification is well understood, translating expert visual judgment into objective, measurable parameters remains challenging. ATC standards are often implicit and experience-driven, making it difficult to formalize them into strict computational rules. |

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| 1.2 Mention the objectives set at the beginning. | **1.2 Objectives:**<br>• To automate the Animal Type Classification process for cattle and buffaloes using AI and image processing.<br>• To reduce subjectivity, bias, and human error present in manual ATC methods.<br>• To accurately detect anatomical landmarks and extract key body structure parameters.<br>• To generate standardized and reproducible ATC scores.<br>• To enable seamless digital integration with the Bharat Pashudhan App.<br>• To support scientific decision-making in breeding programs under PT and PS schemes. | Additionally, ensuring that the AI-generated scores align with nationally accepted breeding program requirements demands careful calibration and validation. There is also a risk that complex physical traits may be oversimplified when reduced to numerical scores, potentially affecting biological relevance. |
| **2. Literature Survey Progress**<br><br>2.1 Summary of at least 10 relevant literature papers (APA/IEEE format).<br><br>2.2 Literature gap identified (how existing work falls short). | **2.1 Summary of Literature:**<br>A total of ten relevant research works were reviewed, prior studies apply computer vision and deep learning for cattle detection, pose estimation, body measurement, behavior analysis, and weight prediction, demonstrating the effectiveness of anatomical keypoint detection and non-contact imaging. However, most works focus on individual tasks rather than a complete animal type evaluation pipeline.<br>**2.2 Literature Gap:**<br>There is a lack of end-to-end, low-cost ATC systems that convert RGB images into standardized classification scores. Existing methods often rely on specialized hardware, ignore field variability, and do not support large-scale or government-level deployment, highlighting the need for a scalable, image-based ATC solution. | The literature review revealed that most existing studies focus on isolated tasks such as cattle detection, pose estimation, or weight prediction rather than a complete end-to-end ATC system. Many works rely on controlled environments or specialized hardware, limiting their applicability in real farm conditions. Furthermore, there is a lack of standardized datasets and evaluation benchmarks for ATC, making objective comparison across studies difficult. Limited field-level validation, especially in Indian livestock settings, highlights a significant research gap. |
| **3. Data Collection & Requirement Analysis**<br><br>3.1 Dataset description, source, and reason for selection. | **3.1 Dataset Description:**<br>The project uses a keypoint-annotated cattle image dataset containing RGB images captured in real farm environments with variations in breed, posture, lighting, and background. Each image is labeled with anatomical keypoints such as shoulder, hip, rump, withers, and leg joints, enabling | Data collection posed significant challenges due to real-world variability in lighting conditions, background clutter, camera angles, and animal posture. Frequent movement of animals and partial occlusion of anatomical landmarks |

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| 3.2 Exploratory Data Analysis (EDA) and reports (graphs, missing values, distributions, correlations, etc.). | extraction of body measurements required for Animal Type Classification (ATC). The dataset is sourced from Roboflow Universe (Cow Keypoint Dataset) and is compatible with pose estimation models like YOLOv8-Pose.<br>**Reason for Selection:**<br>The dataset was selected for its pre-annotated keypoints, real-world variability, compatibility with modern pose estimation architectures, and public availability ensuring reproducibility.<br>**3.2 Exploratory Data Analysis (EDA):**<br>• Analyzed image quality, resolution, and lighting variations<br>• Examined anatomical keypoint distribution and visibility<br>• Identified posture, orientation, and occlusion patterns<br>• Assessed background complexity and annotation consistency | affected keypoint visibility and annotation consistency.<br><br>Variations in breed, size, and orientation further complicated uniform measurement extraction. Since the dataset was not originally designed for ATC scoring, additional assumptions and preprocessing were required to adapt it for the intended objectives. |
| **4. Preliminary Designs / Algorithms**<br>4.1 Initial workflow, block diagram, or conceptual framework showing the proposed approach.<br><br>4.2 Selection and brief justification of intended techniques, algorithms, or models to be used (no implementation yet).<br><br>4.3 Description of planned evaluation metrics and validation approach | **4.1 Initial Workflow / Conceptual Framework:**<br>• Single RGB image capture/upload via mobile or web interface<br>• Image preprocessing and animal detection to isolate the subject<br>• Pose estimation to detect anatomical landmarks<br>• Feature extraction and conversion to standardized ATC scores<br>**4.2 Planned Algorithms / Models:**<br>• Image preprocessing: resizing and normalization<br>• Animal detection: YOLOv8<br>• Pose / landmark detection: YOLOv8-Pose<br>• Scoring: rule-based or weighted ATC scoring model<br>**4.3 Evaluation Metrics:**<br>• Detection performance: Precision, Recall, mAP<br>• Keypoint accuracy: OKS, mean keypoint error<br>• Measurement accuracy: percentage error, MAE<br>• ATC consistency: AI vs expert score deviation | The preliminary design relies heavily on accurate pose estimation, which is sensitive to image quality and annotation precision. Small errors in keypoint detection can propagate and lead to significant inaccuracies in derived body measurements and final ATC scores. The rule-based scoring mechanism may struggle to generalize across different breeds and age groups without extensive tuning. Additionally, achieving a balance between model accuracy and computational efficiency is critical for real-world deployment on low-resource devices. |

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| **5. Progress Status**<br>5.1 Percentage of work completed (overall and for each major stage | • Problem Definition: Completed.<br>• Literature Survey: Completed (10 papers reviewed with references).<br>• Dataset Collection and Preprocessing: Completed (Roboflow dataset finalized).<br>• Preliminary Design / Workflow: Completed (conceptual design and model selection finalized).<br>• Implementation: In progress (YOLOv8 training initiated, migrated to Google Colab for GPU-based training).<br>**Overall Completion : ~30–35%**<br>• Literature review and dataset preparation: 100% complete.<br>• Ladmark Detection : Initiated (~10–15% progress). | The initial implementation phase was delayed due to reliance on CPU-based training, which proved inefficient for deep learning workloads. Migration to GPU-enabled platforms improved training speed but introduced setup complexities and dependency management issues. As landmark detection is still in the early training phase, model accuracy remains unstable. Overall progress has been constrained more by computational limitations than by conceptual or design-related challenges. |

**Supervisor Comments:**

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**Verified by:**

Signature of Supervisor: _____

Date: _____

Signature of Class Teacher: _____

Date: _____